

Parcel-Scale Crop Type Classification Using Tile-Slice-Based Spatial-Temporal Convolutional Networks

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Abstract: Objectives: Parcel-based crop classification using multi-temporal satellite images is essential for precision agriculture. However, exploring multi-scale spatial information for identifying crop types from remote sensing images remains challenging. **Methods:** This paper proposes a tile-slice-based spatial-temporal (TSST) method for parcel-scale crop type classification using multi-temporal Sentinel-2 images. The core idea combines tile-slice-based parcel feature representation with a deep spatial-temporal convolutional network. First, according to crop types and parcels, spatial tiles are sliced to produce a tile-slice-based training dataset and prediction dataset. A deep spatial-temporal convolutional network then estimates crop-type probabilities for the prediction dataset. Finally, the probabilities are fused within parcel polygons to generate the final crop-type maps. **Results:** In the study area in France, parcel-based time-series crop classification using multi-temporal Sentinel-2 images was used to validate the proposed method. Overall accuracy, precision, and F1 score improved by 0.03, 0.02, and 0.02, respectively, compared with the best comparison method. **Conclusions:** TSST is effective for parcel-based crop classification using Sentinel-2 image sequences. Spatial division and slicing are particularly beneficial for larger and spatially concentrated parcels. The TSST model is lightweight but powerful and provides a useful approach for parcel-based remote-sensing time-series analysis.

Key words: Sentinel-2; parcel-scale; crop type mapping; spatial-temporal convolutional networks; spatial division and slicing

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1 Introduction

Remote sensing has advantages including broad coverage, short repeat-observation intervals, limited imaging restrictions, and low cost, and has been successfully applied to regional and global resource surveys, crop growth monitoring, and food-security early warning in agricultural production and management. Remote sensing monitoring and mapping of crop planting types are fundamental to precision-agriculture applications such as crop-growth assessment and yield estimation. However, because crop growth patterns can be similar and a single satellite observation provides only limited information, accurately identifying multiple crop types—especially crops grown in the same season—from a single-date image is difficult. Therefore, exploring and representing the temporal-sequence information contained in multi-temporal remote-sensing images has become an effective approach for crop-type identification and mapping.

For extraction and analysis of multi-temporal remote-sensing information, researchers have proposed statistical models, traditional machine-learning models, and deep-learning models. These approaches focus on time-series techniques such as sequence reconstruction, temporal-component decomposition, temporal smoothing, and temporal classification, improving the performance and application scope of remote-sensing crop-type mapping.

With increasing spatial resolution in Earth observation, satellite imagery can more accurately represent the geometric boundaries of agricultural parcels and high-level information such as within-parcel texture. Consequently, parcel-scale spatial and temporal analysis has become a mainstream approach to precision crop mapping, compared with conventional pixel-based crop classification. Extracting spatial information within crop parcels is important for improving crop-type mapping.

Compared with pixel-based methods using spectral features, object-oriented analysis can extract shape, texture, and contextual relationships from multi-scale image segments, extending the feature dimensions available for crop monitoring and improving mapping accuracy. However, these methods are often constrained by the quality of earlier multi-scale image segmentation, making it difficult to obtain accurate parcel objects and increasing uncertainty in parcel feature representation.

The rapid development of deep learning provides new ideas for agricultural remote-sensing crop classification. Because remote-sensing time-series observations resemble video sequences, existing semantic-segmentation and video-analysis methods can be transferred and extended to crop-type identification. Although these methods can achieve good classification results, they generally rely on ideal labeled samples in which crop labels cover the sample space. This assumption does not match the reality of agricultural field sampling.

Irregular parcels and deep convolutions can also cause crop-feature information to diffuse and become confused near parcel boundaries, reducing parcel-scale classification performance, especially for small parcels. Therefore, researchers have adopted object-oriented image-analysis ideas and divided parcel-scale crop-type recognition into two major stages: parcel extraction and type identification.

For type identification, researchers have built simple spectral-statistical features within parcels or used pretrained convolutional networks to extract multi-level semantic image features, constructing multidimensional temporal parcel features. Deep recurrent networks can then mine short- and long-term dependencies in those temporal features for crop-type identification. Although these approaches outperform many traditional methods, their spatial analysis still relies partly on manually designed high-level spatial features and does not provide an end-to-end, multi-level spatial feature representation model.

Overall, current deep-learning approaches for remote-sensing crop classification still have two limitations: (1) insufficient extraction of high-level spatial information inside parcels, often relying on manually designed spectral and texture features; and (2) a lack of end-to-end deep-network models designed specifically for spatiotemporal crop-type identification.

To address these issues, the paper proposes a tile-slice-based spatial-temporal (TSST) classification network. Its core is parcel-space spatial division and stratified representation combined with a deep spatial-temporal classification network. The framework is designed for parcel-scale spatiotemporal analysis of remote-sensing imagery and provides a lightweight, efficient deep model for crop-type identification. Experiments using Sentinel-2 time-series imagery in a study area in France evaluate the method.

中,训练样本集和验证样本集分别用于调整模型的参数和超参数,测试样本集用于评估分类模型的泛化性能。

2.2 地块关键点与空间分区

空间分区是利用深度学习进行影像时空分析的首要步骤。通过空间分区将大范围影像(实验中Sentinel-2影像的空间尺度约为 $9\,000 \times 9\,000$ 像素)划分为多个影像瓦片,缓解深度学习模型对显存等计算资源的需求;同时空间分区生成的影像瓦片也是深度学习模型的处理对

元的影像区域为该关键点对应的影像瓦片。一个地块多边形可能生成多个关键点,因此其也将对应多个影像瓦片。值得注意的是,影像瓦片尺寸 N 是一个关键参数, N 过大将导致影像瓦片区域内有过多的类别或地块,后续分层处理将产生更高的数据缺失率;而 N 过小会限制深度卷积的感受野,难以捕捉到地块影像的多层次信息。为此,在综合考虑研究区地块尺度、影像空间分辨率以及数据缺失率等因素的基础上,设置 $N=32$ 。

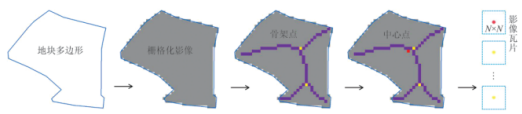


图3 地块关键点提取与影像空间分区示意图
Fig. 3 Key Points of Parcels and Spatial Division of Images

2.3 类别/地块分层

类别/地块分层是利用深度学习进行遥感作物时间序列分析的关键。在影像空间分区中,本文提出的类别/地块分层流程如图4所示。

在基于作物类别的分层中,分别提取影像瓦

片区域内每种类型像元的区域并标记其他像元区域为空白(影像像元值设置为0);在地块分层中,则依据地块编码(如地块序号)提取编码相同的像元区域,并将其他区域像元值设置为0。在构建作物类型识别训练样本中,由于地块类别已知,因此

2.3 Crop-Class / Parcel Stratification

Crop-class/parcel stratification is a key step for deep-learning-based remote-sensing time-series analysis. For crop-class stratification, pixels belonging to each crop type within an image tile were extracted and all other pixels were set to blank (pixel value 0). For parcel stratification, pixels sharing the same parcel identifier were extracted and other pixels were set to 0.

When constructing crop-type training samples, the crop class is known, so each tile region is stratified by class and each stratum generates a labeled sample. When constructing prediction samples, the tile region is stratified by parcel, with each parcel stratum corresponding to one prediction sample. This process reduces interference among different crop classes or parcels and produces relatively “pure” sample features.

Experiments showed that blank regions caused by missing data could make training unstable, prevent convergence, or even cause training failure. Two simulated filling methods were therefore tested: random filling and normal-distribution filling. For random filling, valid pixels were randomly selected and copied into missing locations. For normal filling, valid pixel values were assumed to follow a normal distribution; their mean and standard deviation were calculated, and random values within ± 3 standard deviations were generated to fill missing locations.

Because deep-learning models have some tolerance to missing data, simulated filling was applied only when the average missing rate of an image-tile sequence exceeded 0.6, reducing the missing rate to below 0.6.

2.4 TSST Network Model

Three-dimensional convolutional neural networks can learn joint spatial-temporal information from image sequences. To avoid the large computational cost of direct 3D convolution, and inspired by Pseudo-3D (P3D) and ResNet3D, the authors decomposed conventional 3D convolution into 2D spatial convolution and 1D temporal convolution and proposed the TSST network.

The network contains five submodules: the Spatial Embedding Block (SEB), Dense Spatial Block (DSB), Channel-Spatial Attention (CSA), Vector Block (VB), and a classification block based on a Temporal Convolutional Network (TCN). SEB wraps the input time-series images spatially to generate high-dimensional feature representations. DSB extracts multi-level spatial information. CSA learns attention in the spatial and band/attribute dimensions. VB converts sequence image features into multidimensional temporal information. TCN maps multidimensional temporal features to crop classes and outputs crop-type probabilities.

Figure 4. Parcel slicing based on crop types or parcel coding.

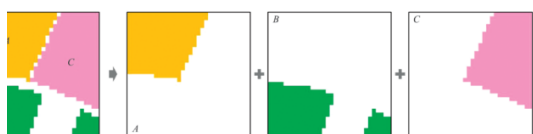


图4 基于作物类别/地块的分层示意图

Fig. 4 Parcel Slice Based on Crop Types or Parcel Coding

片区域进行类别分层,每层生成一个标记以制作预测样本中,则对瓦片区域进行地每个地块分层对应一个预测样本。通过块分层,能够有效排除不同类别或地块之干扰,获取“纯净”的样本特征数据。

过程中也注意到,瓦片分层中的空白区域(缺失)可能会导致模型训练过程不稳定,甚至收敛甚至训练失败等问题。为此,尝试随机填充和正态填充两种方法对空白区域填充,以期改善模型训练过程。在随机从有效像元中随机选取像元,并将其像元

巨大计算资源消耗,受Pseudo-3D (P3D)^[28]、ResNet3D^[29]等三维模型的启发,将常规的三维卷积分解为二维空间卷积和一维时间卷积,提出的TSST网络模型如图5所示。整体上,网络包括空间嵌入模块(spatial embedding block, SEB)、稠密空间连接模块(dense spatial block, DSB)、注意力模块(channel-spatial attention, CSA)、向量化模块(vector block, VB)和基于时域卷积网络(temporal convolutional network, TCN)的分类模块等5个子模块。SEB模块对输入时序影像进行空间包裹生成输入影像的高维特征表达;DSB模

块用于提取影像的多层次空间信息;CSA模块学习影像在空间和波段(属性)维度的注意力;VB模块将序列影像特征转化为多维时序信息;TCN模块实现从多维时序特征到作物类别的映射,并输出作物类别概率。

1) 多层次空间特征

TSST网络模型利用DSB模块提取影像的多尺度空间信息,其实现沿用了DSNet^[30]中的稠密膨胀卷积模块(图5)。具体而言,DSB利用具有密集感受野的膨胀卷积来捕获影像空间的多

4 TSST网络模型

三维卷积神经网络能够学习影像序列的时空联合信息。本文为了避免直接使用三维卷积的

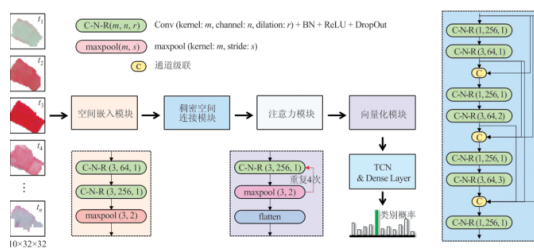


图5 TSST网络模型的整体架构

Fig. 5 Overall Architecture of TSST Network Model

Figure 5. Overall architecture of the TSST network model.

1) Multi-level spatial features. The TSST network uses the DSB module to extract multi-scale spatial information. Its implementation follows the dense dilated-convolution module of DSNet. DSB uses dilated convolutions with dense receptive fields to capture multi-scale spatial information while keeping the number of model parameters relatively low. Dense channel concatenation is used to obtain multi-scale features across image bands, and a 1×1 convolution before the dilated-convolution layers prevents the number of feature channels from growing excessively.

2) Time-series classification. Compared with recurrent neural networks for sequence modeling, TCNs provide multi-level receptive fields, stable gradient propagation, and lower resource consumption. The paper uses TCN to map parcel time-series features to crop types. Dilated convolution captures long-term dependencies between observations, while residual connections transmit information across network layers. The TCN contains three consecutive 1D convolution layers followed by batch normalization, ReLU activation, and dropout. The temporally encoded features are flattened and passed to a fully connected layer with batch normalization, ReLU, and dropout, and Softmax calculates crop-type probabilities.

3) Loss function. To address crop-type class imbalance in the study area, a multi-class Focal Loss was used to train the spatial-temporal classification model.

2.5 Parcel-Scale Crop-Class Merging

Spatial division generates multiple image tiles for a parcel, and each tile produces crop-type probabilities. These probabilities must therefore be fused to obtain the final parcel crop type. Let n_i be the number of valid pixels in prediction tile i that fall within both the tile and parcel polygon, and let $p_i = (p^1_i, p^2_i, \dots, p^T_i)$ be the crop-type probability vector. Inspired by voting in ensemble learning, the method calculates a weighted sum of crop probabilities using n_i as the weight and selects the class with the highest probability as the final parcel crop type.

Equation (1): $P_{type} = \arg \max_{t \in (1,T)} \sum_{i=1}^K n_i \times p^t_i$. Here, P_{type} is the final parcel crop type; K is the number of key points of a parcel; and T is the number of classes in the classification system.

2.6 Performance Evaluation and Comparison

1) Comparison methods. Several methods were used to validate TSST. They were divided into pixel-scale and parcel-scale time-series classification methods. Pixel-scale methods use pixel time-series spectral values as input and employ LSTM and Transformer models to extract temporal features and classify pixels. Parcel polygons are then used as spatial constraints to vote and merge pixel classification results into parcel-scale results. In the parcel-feature approach, the mean spectrum within each parcel is extracted to construct parcel time-series observations, which are then input to LSTM and Transformer models.

For fairness, all models used the same settings for optimizer, loss function, batch size, initial learning rate, early stopping, and related parameters, and the best trained model was used for testing.

2) Evaluation metrics. A confusion matrix was constructed by comparing the reference and predicted class of each test parcel. Overall accuracy (OA), precision (P), recall (R), and F1 score were calculated. OA is the ratio of correctly classified parcels to all test parcels. F1, the harmonic mean of precision and recall, better expresses the classification performance of an individual crop class. Higher OA, P, R, and F1 indicate better classification.

3 Experiments and Discussion

3.1 Experimental Results

1) Crop-type classification results. The TSST network generated a crop-planting-type map and local details, shown in Fig. 6. Visual interpretation was used to evaluate the overall identification effect. Grassland is mainly distributed in the southwest and along both banks of the Vingeanne River (detail C). Grapes are mainly planted in a narrow valley between Chenôve and Beaune (detail B). Winter wheat, winter barley, maize, and other crops dominate the flat eastern-central part of the study area.

2) Accuracy analysis and comparison. Several crop-classification methods were calculated and compared, including pixel-scale LSTM and Transformer methods, parcel-scale class-voting LSTM-V and Transformer-V, and parcel-feature LSTM-P and Transformer-P. The results are shown in Table 1.

Overall, TSST achieved the best crop-classification performance, with OA = 0.8335, P = 0.7187, R = 0.7438, and F1 = 0.7310. Compared with the next-best method, TSST improved OA, P, and F1 by 0.03, 0.02, and 0.02, respectively. Pixel-scale LSTM and Transformer produced the lowest crop-classification accuracy, whereas parcel-scale methods performed better. Using parcel polygons as spatial constraints for feature statistics and crop-type merging can remove outliers and salt-and-pepper noise in feature extraction and classification.

Compared with class-voting and parcel-feature methods, TSST more effectively extracts and uses multi-level spatial information from imagery, resulting in the best classification. The increase in precision was larger than the increase in recall, possibly because high-level parcel spatial semantics were extracted and used, improving identification precision.

Figure 6. Crop-classification results of the proposed TSST network model.

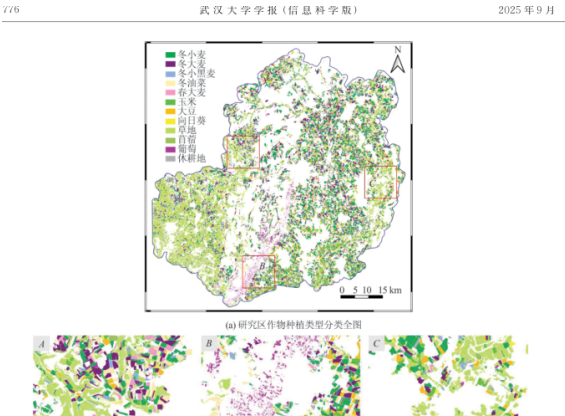


Table 1. Accuracy comparison of crop-type classifications.

Method	OA	P	R	F1
Pixel LSTM	0.7561	0.5457	0.6427	0.5903
Pixel Transformer	0.7693	0.5614	0.6600	0.6067
Parcel class-vote LSTM-V	0.8082	0.6572	0.7445	0.6982
Parcel class-vote Transformer-V	0.7916	0.6732	0.7116	0.6919
Parcel feature LSTM-P	0.8003	0.6582	0.7456	0.6992
Parcel feature Transformer-P	0.8013	0.6946	0.7204	0.7073
TSST	0.8335	0.7187	0.7438	0.7310

3.2 Analysis and Discussion

1) Crop types and accuracy improvement. Using LSTM-P as the baseline, the authors analyzed the F1 scores and improvements for different crop types. TSST and LSTM-P both achieved relatively high accuracy for winter rapeseed, grapes, and grassland, but performed less well for winter triticale, maize, and soybean. Grapes and grassland have relatively distinctive growth processes. Although Focal Loss was used to address class imbalance, classes with more samples still tended to achieve higher classification accuracy.

TSST produced clear improvements for winter rapeseed, maize, and soybean, but slight decreases for winter triticale, grapes, alfalfa, and fallow land. Improvements were generally larger for crops occupying larger, spatially concentrated parcels, whereas small and scattered parcels performed worse. The spatial division and stratification process can increase the missing-value rate in tile sequences and make spatial-feature learning more difficult. Small, scattered crop parcels have higher missing rates, more feature noise, and less stable recognition patterns.

Figure 7. Classification accuracies and improvements for different crop types.

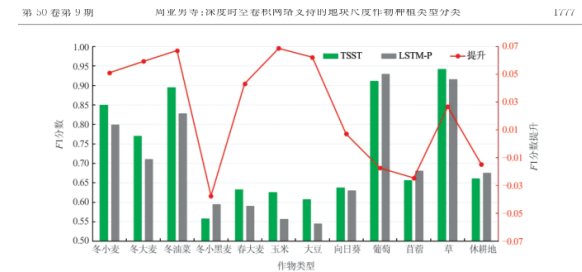


图7 不同类型作物的识别精度及其提升

进一步分析 TSST 网络模型在不同作物上的性能提升,其中在冬油菜、玉米和大豆作物上提升明显,而在冬小黑麦、葡萄、苜蓿和休耕地类别上出现了轻微的精度下滑。从作物地块的平均来看,相对于 No 方案中作物空间特征会受到污染像元的干扰,其他方法均消除了污染像元,并且 Random 和 Normal 方法还模拟填充了缺失值。上述两种填充方法并没有取得较 Zero 处理

2) Cloud/shadow missing-value handling. TSST reduces confusion among spatiotemporal features of different classes through spatial division and class/parcel stratification, but this also produces more missing pixel values in image tiles. Two missing-value reconstruction schemes were tested: random filling and normal-distribution filling. In Table 2, “No” means no cloud/shadow masking, “Zero” means direct zero filling, and “Random” and “Normal” denote the two simulated filling methods. Epoch is the number of training epochs needed for the model to stabilize.

Overall, Zero, Random, and Normal all produced better accuracy than No. Without masking, crop spatial features can be disturbed by contaminated pixels. The other approaches remove contaminated pixels, while Random and Normal additionally simulate missing values. Random and Normal did not outperform Zero, even though they filled missing values, because filling can damage high-level semantic information such as texture. Their major advantage is faster training and convergence, reducing the number of epochs required.

Table 2. Crop-classification accuracy for different missing-value reconstruction methods.					
Scheme	OA	P	R	F1	Epoch
No	0.7579	0.5223	0.6120	0.5636	92
Zero	0.8335	0.7187	0.7438	0.7310	58
Random	0.8251	0.6890	0.7761	0.7299	42
Normal	0.8345	0.7065	0.7610	0.7327	30

3) Model performance comparison. TSST was compared with ResNet-TCN and P3D, two popular deep spatial-temporal analysis networks. ResNet-TCN and P3D used 50-layer and 63-layer backbone networks, respectively. The three models achieved nearly identical overall classification accuracy. However, TSST had substantially fewer parameters and faster prediction, meaning that it achieved comparable crop-type recognition with a smaller and more lightweight model.

Table 3. Performance comparison of spatial-temporal models for crop classification.			
Network	Parameters (MB)	Prediction time (s)	OA
ResNet-TCN	23.68	2.21	0.8341
P3D	24.97	2.43	0.8297
TSST	1.70	1.19	0.8335

4 Conclusion

Fully exploiting the spatiotemporal information of crop planting in imagery is an effective way to improve remote-sensing crop-type classification and mapping. Based on parcel-space spatial division and stratified representation and a deep spatial-temporal convolutional classification network, this paper proposes a spatially divided and stratified remote-sensing spatiotemporal crop-type classification method, TSST.

Experiments for parcel-scale crop-type mapping in the French study area using Sentinel-2 time-series imagery show that: (1) the TSST network can effectively learn multi-scale spatial-temporal semantic features and improve remote-sensing spatiotemporal crop identification; compared with existing methods, OA, P, and F1 improved by 0.03, 0.02, and 0.02, respectively; (2) spatial division and stratification designed for parcel feature representation are more beneficial for identifying crop types occupying large, spatially concentrated parcels; and (3) TSST is a lightweight and efficient crop-type identification model.

These experiments and conclusions provide new ideas and methods for agricultural remote-sensing time-series analysis. The methods can also be transferred and adapted to other satellite datasets and applications.

Clouds, shadows, and other imaging conditions limit the spatiotemporal observation of crop growth using optical remote sensing. Future research will consider introducing and fusing microwave observations, such as Sentinel-1 and GF-3 satellite data, to improve time-series monitoring and mapping of crop growth.

Reference note: The reference list below is translated/standardized into English while preserving the references contained in the supplied PDF.

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